



Seamless Knitting

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Background

1.1 Context

Seamless knitting represents a highly integrated textile production technology where entire garments are produced directly on knitting machines, removing almost all cutting and sewing operations.

This innovation, (*in this case we base our model on the machines from **Shima Seiki (WHOLEGARMENT®)** and **Santoni (SM8 series)***), achieves near-zero fabric waste and higher material efficiency compared to traditional knitwear production.

However, the process is more energy-intensive per kilogram of fabric due to complex knitting paths, longer cycle times, and higher frictional resistance, particularly with fine yarns.

Average machine characteristics (literature and datasheets):

- **Machine weight:** 750 - 1,400 kg
- **Installed power:** 1.5 - 2.2 kW (varies with gauge)
- **Knitting speed:** 1.3 - 1.6 m/s
- **Production rate:** 0.7 - 2.0 garments h⁻¹
- **Energy range:** ~0.9 - 17 kWh kg⁻¹ garment

(Sources: Shima Seiki 2025; Knox 2011; Maity 2021; NTNU 2024)

1.2 Scope and Functional Unit

The seamless knitting model includes:

- **Direct electricity consumption** of the machine;
- **Indirect energy use** (compressors, HVAC, lighting, automation);
- **Lubricant use** and air emissions;
- **Capital goods and end-of-life allocation**;
- **Material losses (yarn waste)** during operation.

The **functional unit** is **1 kg of knitted garment** produced under average industrial conditions.

Methodological Overview

2.1 Yarn losses and lubricant - dependency on yarn fineness (dtex)

Concept

Yarn losses and lubricant demand depend on yarn fineness and machine efficiency.

Finer yarns require longer running time, higher tension control, and more lubricant per kilogram of output.

Base definition

"Input quantity refers to the net yarn required to produce 1 kg of knitted fabric. Yarn losses during the knitting process (thread breaks, edge waste, trial runs) should be included."

(Laursen et al., 2007; van der Velden et al., 2014; Sandin et al., 2019 – Mistra)

Empirical references

- **Material realisation:** 94 - 98 % (Maity et al., 2021; Rakib Ahmed 2024)
- **Lubricant:** 8 g kg⁻¹ fabric "Sandin et al.2019 pp45 - The lubricant was assumed to correspond to 8% of the weight of the knit."

Values applied

Parameter	Average	Comment
Yarn loss	2.5 %	Derived from 94–98 % yarn utilisation
Lubricant	0.008 kg kg ⁻¹	Upper-bound value adopted for high-load operation

2.2 Electricity consumption – direct and indirect energy

Direct electricity (machine operation)

Based on EcoBalyse, Maity et al. (2021) and Shima Seiki datasheets, seamless knitting machines typically consume **1.2–2.1 kWh kg⁻¹** of fabric.

A mid-range **average value of 1.5 kWh kg⁻¹** is adopted for modelling.

Indirect energy (compressors, HVAC, lighting, cleaning)

The same structure used in flat and circular knitting is retained, totalling **4.46 kWh kg⁻¹** distributed as:

- 35 % knitting machines
- 25 % compressors
- 30 % HVAC
- 6.5 % lighting
- 3.5 % auxiliaries

This ensures methodological alignment across knitting technologies.

3. Capital Goods and End-of-Life

3.1 Machine characteristics

Machine	Weight (kg)	Source
Santoni SM8-TR1S	750	Santoni 2023 - SM8-TR1S datasheet

3.2 Lifetime parameters

Scenario	Years	Hours/day	Days/year	Productivity (kg h ⁻¹)	Allocation (kg/kg garment)
Best case (BCS)	20	24	365	26.2	0.0002
Average	15	16	300	26.2	0.0004
Worst case (WCS)	10	8	300	26.2	0.0012

3.3 End-of-life allocation

Material	Share (%)	Allocation (kg/kg garment)	Notes
Steel	85 %	0.0003	Recycled or disposed depending on scenario
Other materials	15 %	0.0001	Includes electronics and polymers

Machine composition:

Approx. 85 % steel (637.5 kg) and 15 % other materials (112.5 kg).

3.4 Building Infrastructure

The seamless knitting process model includes an optional building infrastructure (pavilion) share as a modular and auditable input. The building is represented by an ecoinvent-style construction dataset ("building, hall") whose reference flow is m² of industrial hall.

Allocation principle (area amortisation over lifetime output).

Building infrastructure is attributed to 1 kg of knitted garment by amortising the allocated hall area over the lifetime production associated with the knitting cell:

$$\text{Building intensity (m}^2\text{/kg)} = \frac{A_{\text{alloc}} \text{ (m}^2\text{)}}{Q_{\text{lifetime}} \text{ (kg)}}$$

$$Q_{\text{lifetime}} = \text{productivity (kg/h)} \times \text{hours/day} \times \text{days/year} \times \text{years}$$

where A_{alloc} is the hall area allocated to the knitting cell (machine footprint + access + yarn storage + maintenance space + internal logistics), and Q_{lifetime} is the total knitted output produced during the reference service life under the chosen operating regime.

Technology rationale.

Seamless knitting (body-size knitting on large-diameter circular machines) produces complete garment tubes in a single operation, eliminating cut-and-sew waste. Its throughput per unit of floor space is intermediate between circular fabric knitting and flat or 3D knitting, as the larger machine envelope and more complex programming cycles reduce the production rate per m² relative to conventional circular knitting.

Practical implementation.

In the seamless knitting foreground unit process, we add an input: *building, hall* (unit = m²) with amount = 1.61E-05 m²/kg.

End-of-life of the building.

The building dataset includes construction, maintenance and decommissioning within the background system model. No separate building EoL module is created in the foreground; only the amortised m²/kg input is provided.

Default value (this study). In the absence of site-specific A_{alloc} and Q_{lifetime} , the following reference intensity is applied, shared with the hosiery technology group given comparable machine scale and operating throughput:

Seamless knitting (garment): **1.61E-05 m²/kg**

Physical plausibility was cross-checked using machine envelope data from Santoni SM8-TR1S (A = 2700 mm, D = 3150 mm, envelope ≈ 8.5 m²) with a cell factor of 3–4×. This value also aligns directly with the benchmark reported in the openLCA Nexus organic cotton sweater case study.

4. Waste and Material Efficiency

Traditional cut-and-sew processes typically lose **10 - 25 %** of fabric as trims (AATCC → GIZ 2023).

Seamless knitting eliminates most cutting operations and achieves **> 95 % material utilisation**, corresponding to **< 5 % total waste** (K&G Garment 2025).

Accordingly, waste-to-treatment in the LCI is set at **0.025 kg textile waste per kg garment**.

5. Summary of Inventory Parameters (Average Case)

Flow	Amount	Unit	Comment
Product	1	kg	Average seamless knitted garment
Yarn losses	0.025	kg	Average 2.5 % material loss
Lubricant	0.008	kg	Upper-bound for high-load production
Electricity mix	1.5	kWh	Direct machine operation
Indirect energy	4.46	kWh	Compressors, HVAC, lighting, auxiliaries
Capital goods	0.0004	kg	Machine allocation (average)
EoL steel	0.0003	kg	Machine end-of-life
EoL others	0.0001	kg	Machine end-of-life
Textile waste	0.025	kg	Process and start-up losses

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